Paired single-cell transcriptome and TCR-repertoire analysis reveals convergent CD4+ T cells in a mucosal inflammation model

Sharma, Apoorva; Olmedo, V; Alvarez Rivas, C.

Department of Oral Biology, School of Dental Medicine, University at Buffalo, Buffalo, NY, USA
as587@buffalo.edu

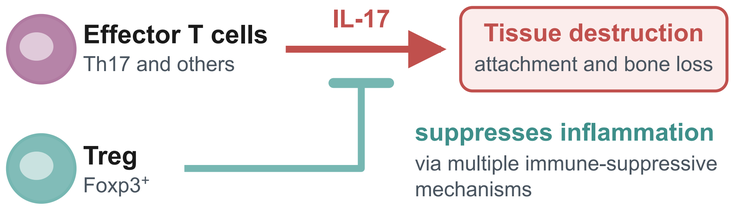
Introduction

Immune regulation dips, then resets
Stress, trauma, infection, hormonal shifts.

What does that leave behind?
Which T cell states expand, which clones persist, and do they drive recurrence?

Periodontitis — a mucosal inflammation model
Chronic, recurrent; destroys tooth-supporting tissue.

In this tissue, T cells hold the balance



Hypothesis

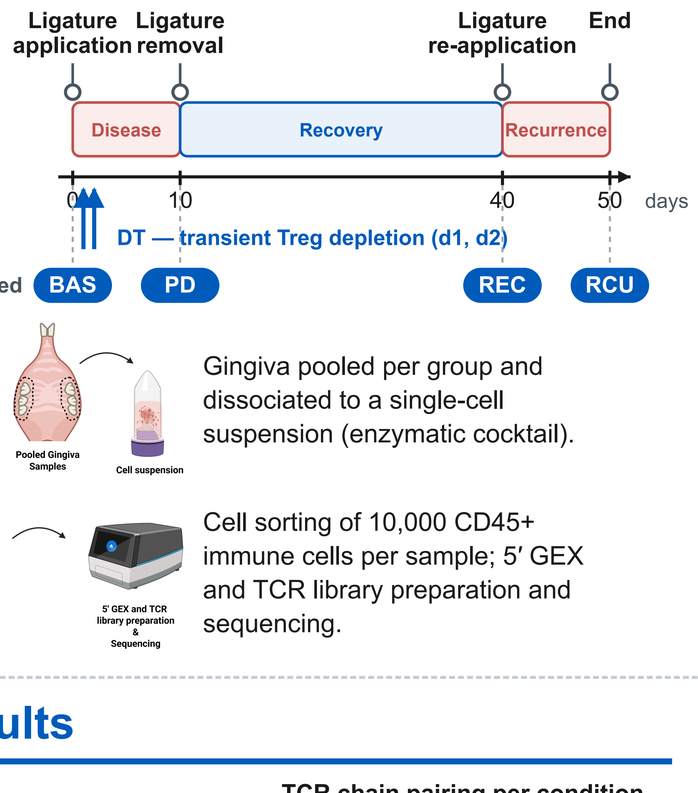
In a transient regulatory disruption model, clonally expanded T cells persist in the mucosa after recovery as a tissue-adapted population, and equivalent clonotypes are re-selected at recurrence.



Methods

SeRPD model
Four experimental groups: BAS (baseline), PD (periodontitis), REC (recovery), RCU (recurrence).

Mouse
Foxp3-DTR mice express the human DTR only in Tregs, making them sensitive to DT.



Gingiva pooled per group and dissociated to a single-cell suspension (enzymatic cocktail).

Cell sorting of 10,000 CD45+ immune cells per sample; 5′ GEX and TCR library preparation and sequencing.

Results

A

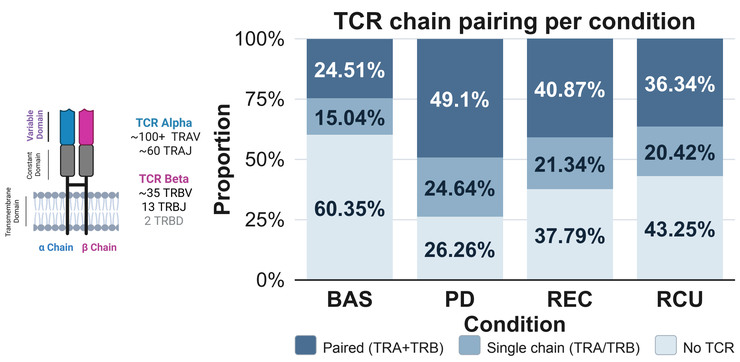


Fig 1. TCR chain pairing per condition — paired (TRA+TRB), single chain, or no TCR, averaged across the two replicates per group.

B

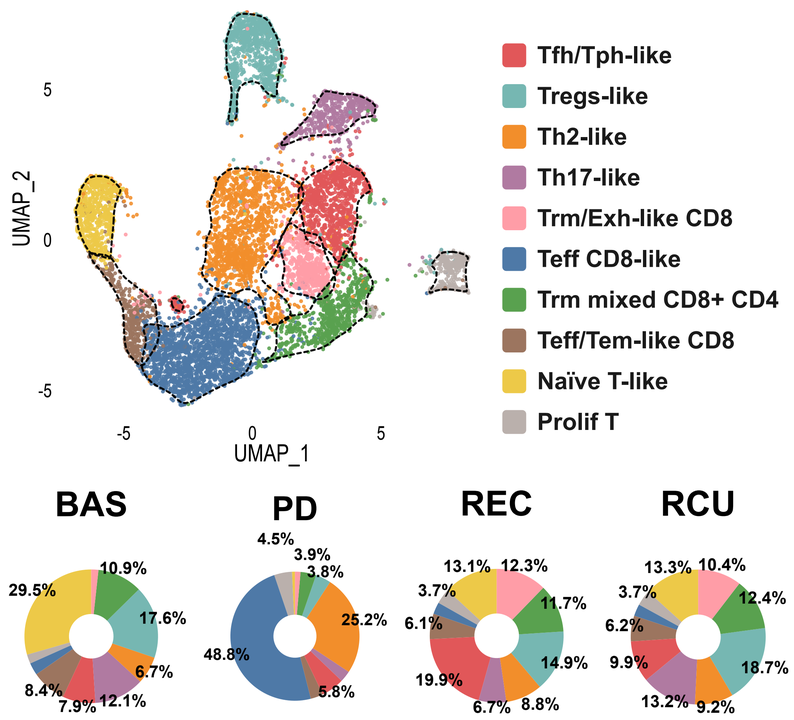


Fig 2. Ten TCR-paired T cell states resolved from paired GEX and TCR, and their proportions in each group.

C

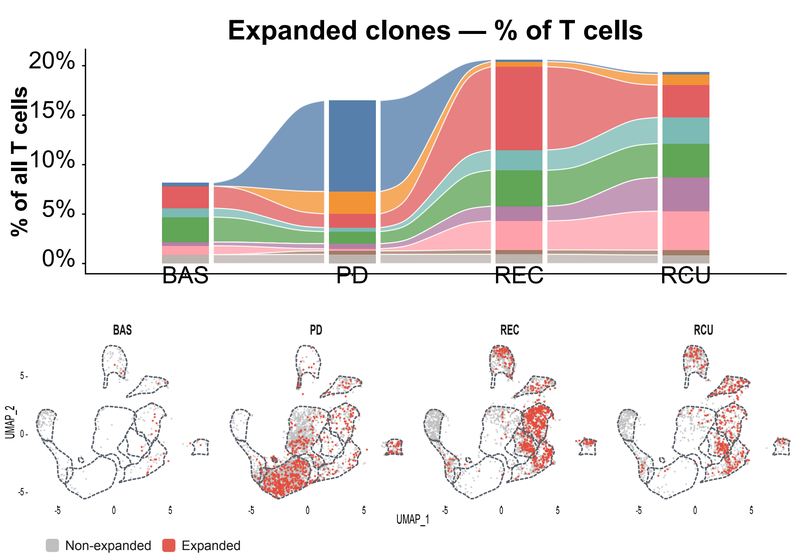


Fig 3. Clonally expanded T cells by cell type across phases (stream), and their position on the T cell UMAP per group (red = expanded).

D

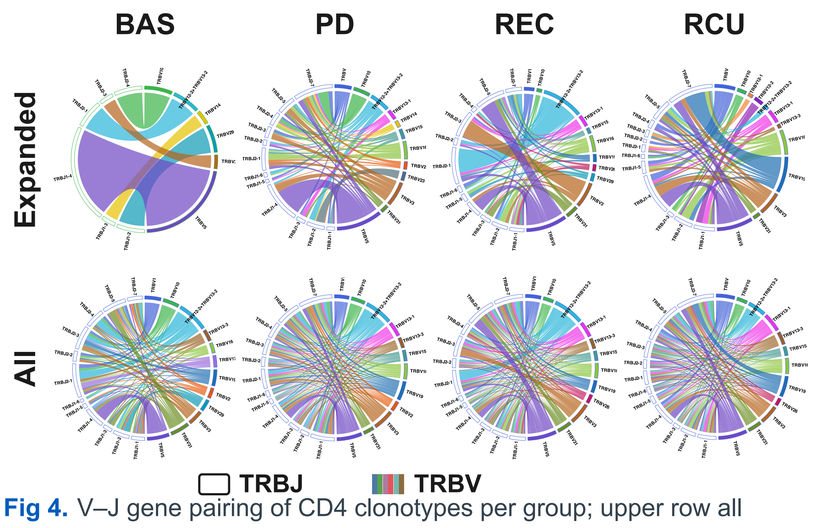
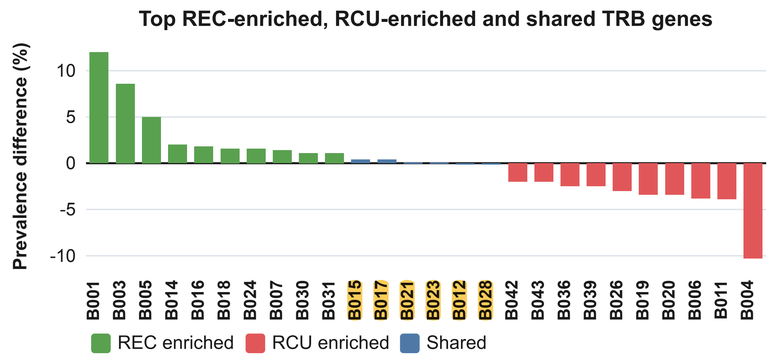


Fig 4. V–J gene pairing of CD4 clonotypes per group; upper row all CD4 clonotypes, lower row expanded clonotypes only.

The expanded compartment keeps a biased subset of pairings rather than sampling the whole repertoire. Some of those pairings - TRBV5 with TRBJ1-4, for example, reappear at both recovery and recurrence.

E



Clonotype overlap between REC and RCU

CD4 clusters	n REC	n RCU	Intersection	Union	Jaccard
Tfh/Tph-like	63	23	10	76	0.132
Tregs-like	34	23	5	52	0.096
Th17-like	17	21	2	36	0.056
Th2-like	11	10	1	20	0.050
Global	**90**	**60**	**25**	**125**	**0.200**

Fig 5. Top REC- and RCU-enriched and shared TRB genes, ranked by prevalence difference between recovery and recurrence.

F Convergent clonotype (highly expanded)

Chain	Genes	CDR3 aa	CDR3 nt
alpha (TRA)	TRAV6N-7.TRAJ13.TRAC	CALGANSGTYQRF	TGTGCTCTGGGGGCAAATT CTGGACTTACCAGAGGTTT
beta (TRB)	TRBV5.TRBD1.TRBJ1-4.TRBC1	CASSQEGQGT NERLFF	TGTGCCAGCAGCCAAG AGGGACAGGGGACCAA CGAAAGATTATTTTTC

Fig 6. **A single alpha/beta clonotype found highly expanded in both recovery and recurrence.**

G

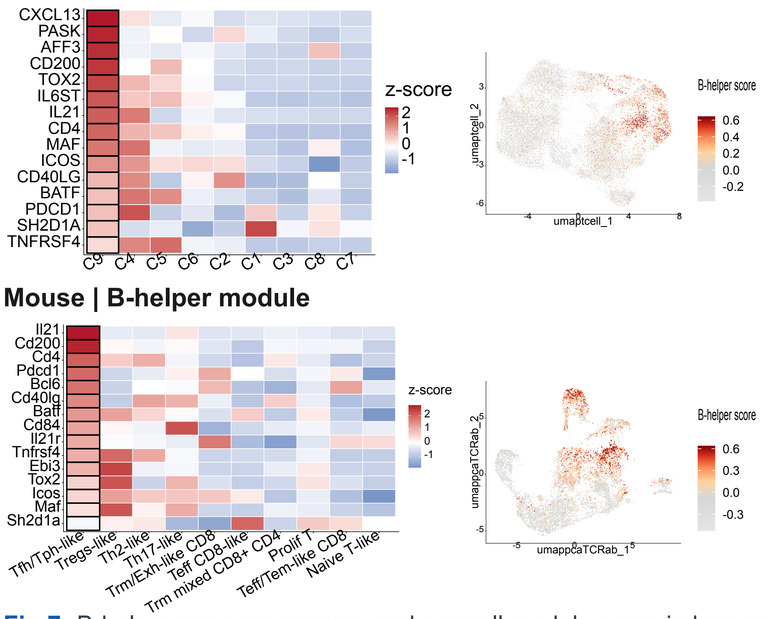


Fig 7. B-helper gene programme and per-cell module score in human and mouse; the Tfh/Tph-like cluster scores highest in both species.

Conclusion

Memory and persistence
Recovered gingiva is not naïve: it retains an enriched CD4+ population, predominantly Tfh/Tph-like.

Clonal convergence
Beta-chain repertoires overlap from recovery to recurrence (Jaccard 0.2), and one αβ clonotype is shared.

Clinical insight
The mouse Tfh/Tph-like cluster shares a B-helper programme with human C9 — a candidate target for prevention.